# MicroBioAgent: an AI agent for automated microbiome analysis

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**Abstract**

Microbiome studies require analytical choices that depend on the biological question, available assays, and intermediate findings. Translating these choices into executable analyses requires connecting heterogeneous methods, software environments, reference resources, and intermediate outputs. Here we present MicroBioAgent, a domain agent that translates biological questions into study-specific analyses across 16S amplicon, reference-based shotgun, and assembly-based metagenomic workflows. MicroBioAgent uses microbiome-specific skills to select and execute analytical methods in user-controlled computational environments. It evaluates intermediate results to determine subsequent analyses in line with the biological question. We evaluated MicroBioAgent on three published datasets through amplicon and shotgun analysis of tomato rhizosphere communities, taxonomic and functional profiling with classifier training in a colorectal cancer cohort, and metagenomic assembly and MAG generation from a model polyethylene degradation study, followed by taxonomic profiling and annotation of genes associated with alkane and oxidized-derivative degradation in focal MAGs. The reanalyses identified biological patterns consistent with the original studies, with differences in statistical support and taxon detection across analytical methods and reference databases. The system also retains the analysis context across long-running operations, allowing pending work to be revised and execution to continue after tool completion. These applications demonstrate that MicroBioAgent can carry out upstream processing and downstream analysis across different microbiome study designs without requiring a fixed end-to-end workflow template.

## 1 Introduction

Advances in sequencing have expanded opportunities to investigate microbial communities in human, plant, and environmental systems. Analyses of 16S rRNA gene amplicons and shotgun metagenomes can characterize community composition, reveal functional potential, and recover microbial genomes [1]. Turning these data into biological insights, however, requires researchers to connect sequence processing with taxonomic and functional annotation and

downstream statistical analysis. Selecting suitable methods and reference databases, configuring software, writing analysis code, and interpreting the results require both computational skills and biological knowledge. These demands create substantial barriers for researchers without specialist bioinformatics training, limiting their ability to conduct and extend microbiome analyses independently [2, 3].

Existing analysis platforms and workflow tools address parts of this burden. Workflow managers such as Snakemake and Nextflow organize computational dependencies and support reproducible, scalable execution [4, 5]. Once an analysis has been specified, these systems can coordinate its computational steps. Researchers must still determine which analyses address their question, configure the required resources, and interpret the outputs. Adding a new comparison or following up an intermediate finding can also require changes to the workflow or additional code. Reducing this burden therefore requires support for planning and revising analyses as well as executing them.

Large language model agents offer a way to connect these decisions with computational execution through natural-language interaction. AutoBA combined analysis planning, code generation, execution, and automated repair for multi-omics tasks, while Biomni integrated tools and resources for a broad range of biomedical analyses [6, 7]. BioMedAgent, BioMaster, and M3A have further explored automated bioinformatics and human supervision of multi-step analyses [8–10]. For microbiome researchers, the practical need is to carry a study objective through an entire analysis, from processing sequencing reads to comparing communities or examining the functions of individual organisms. Meeting this need requires the agent to select appropriate methods for the input data and connect their outputs to subsequent analyses without requiring users to write commands or assemble the workflow themselves.

MetaClaw is an AI agent designed specifically for microbiome analysis. It connects predefined upstream workflows on FlowHub with customizable downstream analyses through a YAML pipeline registry [11]. This architecture brings upstream processing and downstream interpretation into one agent-guided analysis, while retaining registered workflows and an external platform for upstream metagenomic processing. During execution, MetaClaw manages long-running tasks in the background and monitors their progress through repeated status checks within the active conversation. For researchers who need to combine methods beyond registered workflows or process sequencing data on their own infrastructure, flexible composition and local execution of both upstream and downstream analyses remain important requirements. Flexibility during execution also depends on whether users can revise pending analyses while a computation is running. Keeping the conversation occupied with status checks limits this interaction, motivating an execution mechanism that allows background computation and user

feedback to proceed independently.

Here we present MicroBioAgent, an AI-agent framework that translates research objectives into locally executable microbiome analyses. It combines individual analytical skills for 16S amplicon analysis, reference-based metagenomics, and assembly-based metagenomics with downstream statistical analysis and functional annotation, without requiring a predefined end-to-end workflow template. An analysis plan is established before execution and revised in response to intermediate findings or user feedback. Configurable software environments and reference resources support local upstream and downstream execution without uploading raw sequencing reads to an external analysis platform. During long-running computations, the task continues in the background while the conversation remains available for user feedback. The agent can update pending analyses and automatically continue execution after the background task finishes. We evaluated its analytical capabilities through reanalyses of three published datasets comprising 230 sequencing libraries, examining completion of the requested analyses and recovery of reported biological patterns despite differences in methods and databases. We also designed robustness experiments across three language models, three levels of prompt detail, and three repeats per combination (27 sessions), together with component ablations assessing task completion, method adherence, and token consumption. Within the ablation experiments, a comparison with and without the runtime patch specifically examined whether users could revise pending analyses while a background task was running and whether the agent automatically resumed analysis after the task finished. MicroBioAgent aims to make microbiome analysis more accessible to researchers without specialist bioinformatics training by reducing manual coding, tool integration, and supervision of long-running computation while preserving flexibility in method selection and user control.

# 2 Results

## 2.1 Composing study-specific analyses across microbiome workflows

MicroBioAgent is an AI-agent framework that integrates flexible composition of upstream and downstream microbiome analyses, local execution, and asynchronous human-in-the-loop interaction (figure 1). Given a research question expressed in natural language, it develops an analysis plan and selects microbiome skills to carry it out. These skills provide method-specific guidance that the agent draws on when choosing analytical approaches, preparing commands, and interpreting outputs. As each step completes, the agent examines its results and decides whether the pending plan needs adjustment.

MicroBioAgent executes upstream processing and downstream analyses

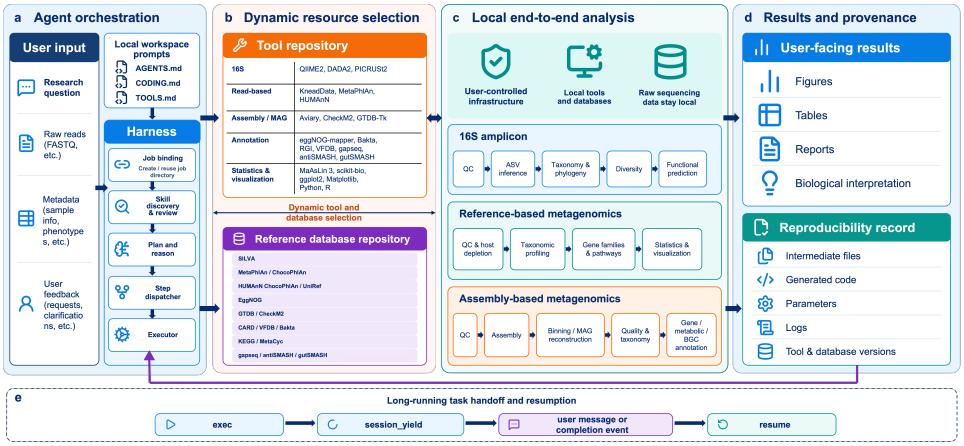



Figure 1: MicroBioAgent architecture and operating model. A user request is bound to an analysis job, from which the agent plans, selects skills, and resolves local runtimes and reference resources for 16S amplicon, reference-based shotgun, and assembly-based metagenomic analyses. Intermediate artifacts, commands, parameters, and logs remain associated with the job. During long-running computation, a user message can update pending work, whereas command completion triggers output verification and continuation in the same session.

in a user-controlled local computational environment, without requiring raw sequencing data to be uploaded to an external analysis platform. To support these analyses, the system manages analytical tools and reference databases through separate configuration files. The tool configuration specifies available software and its associated execution environments, while the database configuration records reference resources and their local storage paths. Users can modify these configurations to add or replace tools and reference databases according to their research needs.

For upstream analysis, MicroBioAgent supports 16S amplicon processing, reference-based metagenomic profiling, and assembly-based metagenomic reconstruction (table 1). Its amplicon capabilities include sequence quality assessment, denoising, amplicon sequence variant inference, taxonomic classification, and phylogenetic reconstruction. For shotgun data, it supports read quality control and host-read removal, taxonomic and functional profiling, and gene-catalogue construction. Assembly-based capabilities include metagenomic assembly and binning, MAG quality assessment, dereplication, taxonomic assignment, and abundance estimation.

MicroBioAgent supports a range of downstream analyses, such as comparisons of microbial communities and characterization of the functions of individual organisms. These include comparisons of diversity and taxonomic or functional abundances between groups, covariate association testing, func-

tional enrichment, association network analysis, and supervised classification. For genes and recovered genomes, available analyses include functional and pathway annotation, antimicrobial-resistance and virulence-gene annotation, metabolic reconstruction, and identification of primary metabolic and secondary-metabolite biosynthetic gene clusters. For visualization, MicroBioAgent supports ordination plots of community differences and heatmaps of taxonomic abundances or functional profiles, with customization to highlight the comparisons of interest.

During long-running computations, such as MetaPhlAn or HUMAnN profiling, MicroBioAgent keeps the conversation available for user interaction while the analysis runs in the background. Users can provide additional instructions or revise the pending analysis plan without waiting for the running task to finish. This asynchronous human-in-the-loop interaction allows the agent to incorporate feedback during execution and automatically resume the analysis under the updated plan after checking the completed task's outcome. MicroBioAgent thus supports end-to-end microbiome analysis through flexible workflow composition and local execution, while allowing users to guide ongoing analyses.

Table 1: MicroBioAgent analytical capabilities, grouped by scientific task. Functional predictions and genome annotations describe inferred biological potential.

| Task category | Supported analyses |
|---|---|
| Amplicon characterization | Read import and quality summaries, denoising and ASV inference, taxonomy, phylogeny, predicted functional profiles |
| Shotgun profiling | Read quality control and host depletion, species abundances, gene-family and pathway profiles, nonredundant gene catalogues |
| MAG characterization | Assembly and binning, quality assessment. Dereplication, taxonomic assignment, abundance estimation, genome annotation |
| Community comparisons | Alpha and beta diversity, ordination, group comparisons, differential abundance, covariate association testing |
| Functional interpretation | Gene and protein annotation, orthology, pathway, and ontology mapping, functional enrichment, resistance and virulence gene annotation |
| Metabolic characterization | Metabolic pathway and model reconstruction, primary metabolic gene clusters, secondary-metabolite biosynthetic gene clusters |
| Study-specific analysis | Custom statistical scripts, association networks, supervised classification, figures, result tables, and analytical reports |
| Data and resource access | Public sequencing metadata and reference-protein retrieval, local reference resources and online annotation queries |

## 2.2 Study-specific analyses across plant, human, and environmental microbiomes

We evaluated MicroBioAgent on three published datasets comprising 230 sequencing libraries from plant, human, and environmental microbiomes (Supplementary Table S1). These case studies assessed the system's ability to complete the requested analyses and recover biological patterns reported in the source studies. Because software versions, reference databases, and some statistical procedures differed, exact computational reproduction was not the evaluation objective. Instead, we assessed completion of the requested analyses, validity of the resulting outputs, and recovery of prespecified biological signals.

The three cases were selected to test MicroBioAgent on distinct combinations of data types and analytical tasks. The tomato rhizosphere case combined 60 16S amplicon libraries with 36 matched shotgun libraries to examine differences in microbial diversity and taxonomic composition associated with pathogen inoculation and cultivar resistance. The colorectal cancer case used 128 faecal shotgun metagenomes from 74 patients and 54 controls to compare species and functional abundances and train classifiers based on gene-family and KEGG Orthology profiles. The third case used six shotgun metagenomes from soil-derived enrichment cultures grown on low-molecular-weight polyethylene or alkane substrates to generate MAGs through assembly and binning, dereplicate them, characterize their taxonomic composition, and annotate degradation-associated genes in focal genomes.

### 2.2.1 A tomato rhizosphere bacterial-wilt resistance study

We applied MicroBioAgent to a published study of tomato rhizosphere microbiomes associated with resistance to bacterial wilt [12]. This case evaluated its ability to connect amplicon and shotgun analyses to examine resistance-associated community changes and identify the species markers reported in that study. The input comprised 60 16S V3–V4 amplicon libraries and 36 matched shotgun libraries from two susceptible cultivars (A57 and HG70), a resistant parent (HG64), and a resistant F1 hybrid (HF12). The amplicon data covered five sampling conditions with three biological replicates per cultivar, including before inoculation (T1), and inoculated and uninoculated plants at day 5 (T2 and T2C) and day 10 (T3 and T3C). The matched shotgun libraries covered T1, T2C, and T2. Starting from sequencing reads, MicroBioAgent performed quality control, DADA2 denoising and taxonomic classification against the SILVA database for the amplicon data, and quality control, host-read removal and MetaPhlAn3 taxonomic profiling for the shotgun data. The resulting profiles supported community-level comparisons using 16S data and species-level marker analysis using shotgun data.

For the 16S amplicon data, MicroBioAgent examined how pathogen inoculation and cultivar resistance were reflected in microbial diversity and community composition. Following *Ralstonia* inoculation, Shannon diversity declined in susceptible cultivars and remained higher in resistant cultivars at both day 5 ($P = 0.002$) and day 10 ($P = 0.004$, figure 2a). Community composition also differed between inoculated and uninoculated samples, which separated along the first weighted-UniFrac PCoA axis (56.3% of variance). Resistant and susceptible samples showed a less pronounced separation along the second axis (12.2%), with overlap between the two groups (figure 2b). Genus-level profiles linked these community changes to an expansion of *Ralstonia*, most pronounced in the susceptible cultivars A57 and HG70 (figure 2c). These patterns were consistent with the inoculation-associated community changes and higher post-inoculation diversity in resistant cultivars reported in the source study.

To examine the taxa underlying these resistance-associated differences, MicroBioAgent compared each resistant cultivar with the pooled susceptible cultivars using MaAsLin 3. In the 16S profiles, the number of genera enriched in both HG64 and HF12 ranged from 6 to 15 across sampling conditions, with different genera contributing to these shared enrichments (Supplementary Figure S1). The matched shotgun data provided species-level comparisons of the markers highlighted in the source study. MetaPhlAn3 profiles showed enrichment of the *Pseudomonas putida* group in both the resistant parent HG64 ($P = 0.013$) and the resistant hybrid HF12 ($P = 0.009$) at T1. The other reported marker, *Sphingopyxis unclassified*, was not identified in the MetaPhlAn3 profiles. Following investigator review, MicroBioAgent repeated the shotgun profiling using MetaPhlAn2, the version used in the original study, and identified both *Pseudomonas putida* and *Sphingopyxis unclassified* as enriched in resistant cultivars, recovering both reported taxonomic associations.

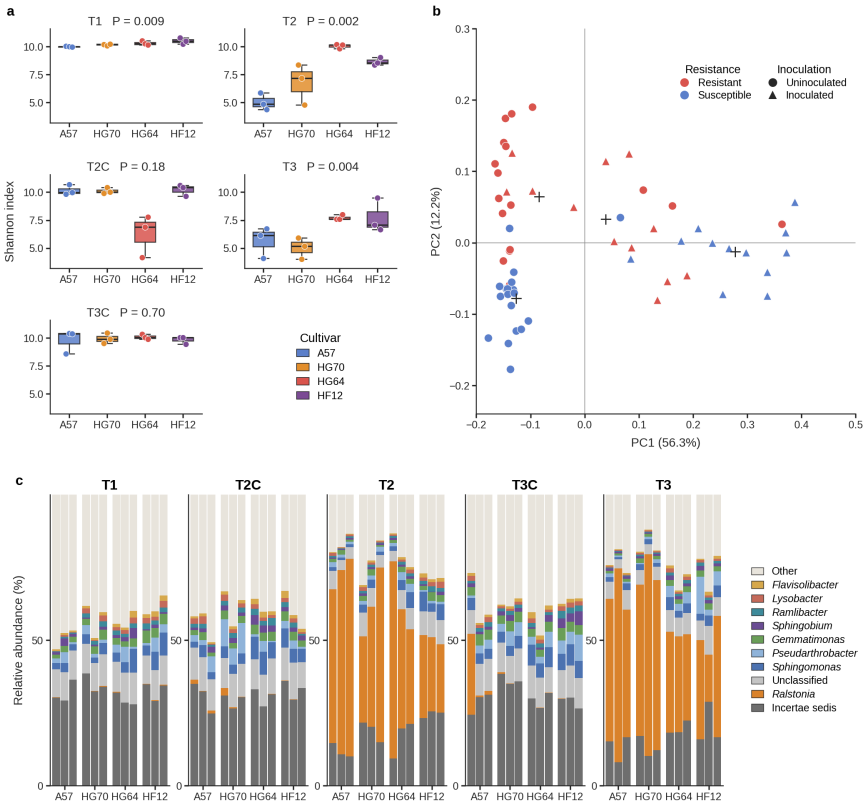



Figure 2: Tomato rhizosphere diversity and community composition. (a) Shannon diversity across cultivars, sampling time points, and inoculation conditions. (b) Weighted-UniFrac PCoA, coloured by resistance phenotype and shaped by inoculation treatment. (c) Genus-level composition across individual samples. Each cultivar-time-point combination contained three biological replicates.

### 2.2.2 A colorectal cancer faecal metagenomic biomarker study

The colorectal cancer (CRC) case evaluated whether MicroBioAgent could connect taxonomic and functional profiling to disease-associated species analysis, co-occurrence networks and predictive modelling. The dataset comprised 128 faecal metagenomes from the Hong Kong C1 cohort reported by Yu *et al.*, including 74 patients with CRC and 54 controls [13]. MicroBioAgent performed species-level profiling with MetaPhlAn3 and quantified UniRef90 gene-family and KEGG Orthology (KO) abundances with HUMAnN3 for downstream association analysis and classification (figure 3). Reported species associations and co-occurrence patterns provided biological reference points, while cross-validation assessed the predictive performance of the functional features.

PCoA based on Bray–Curtis dissimilarities calculated from MetaPhlAn3 species-level relative abundance profiles showed no clear separation among controls, early-stage CRC cases ($n = 36$) and late-stage CRC cases ($n = 38$). PERMANOVA nevertheless detected differences in community composition associated with CRC status ($F = 2.32$, $P = 0.002$) and disease stage ($F = 1.65$, $P = 0.009$). Separate tests for sex, age and BMI were not significant.

Comparisons of species abundances between cases and controls identified eight CRC-enriched and two control-enriched taxa after Benjamini–Hochberg correction ($q < 0.05$). Four CRC-enriched species, *Parvimonas micra*, *Peptostreptococcus stomatis*, *Fusobacterium nucleatum* and *Solobacterium moorei*, showed the same direction of association as in the original study. Enrichment of *Eubacterium ventriosum* in controls also agreed with the reported association. To examine whether these disease-associated species also varied together across samples, Spearman co-occurrence analysis was performed on the ten differentially abundant species. The network contained 16 positive correlations ($|\rho| \geq 0.4$, BH-adjusted $q < 0.05$), with *Parvimonas micra* connected to six other species. Its positive correlations with *Fusobacterium nucleatum* ($\rho = 0.49$) and *Solobacterium moorei* ($\rho = 0.53$) were consistent with the co-occurrence relationships emphasized in the original study [13].

The analysis then extended from identifying disease-associated taxa to testing whether functional differences could distinguish CRC cases from controls. MicroBioAgent selected differentially abundant features and trained separate gene-family and KO classifiers to compare the predictive performance of these two functional representations. Within each training fold, mRMR selection yielded 15 gene-family features or 15 KO features for logistic-regression modelling. Performance was evaluated by nested fivefold cross-validation within the C1 cohort. The gene-family classifier achieved a pooled out-of-fold AUROC of 0.902, compared with 0.632 for the KO classifier, indicating stronger discrimination using gene-family features. At thresholds selected using Youden's index, the gene-family classifier achieved a sensitivity

of 0.82 and specificity of 0.89, compared with 0.41 and 0.87 for the KO classifier.

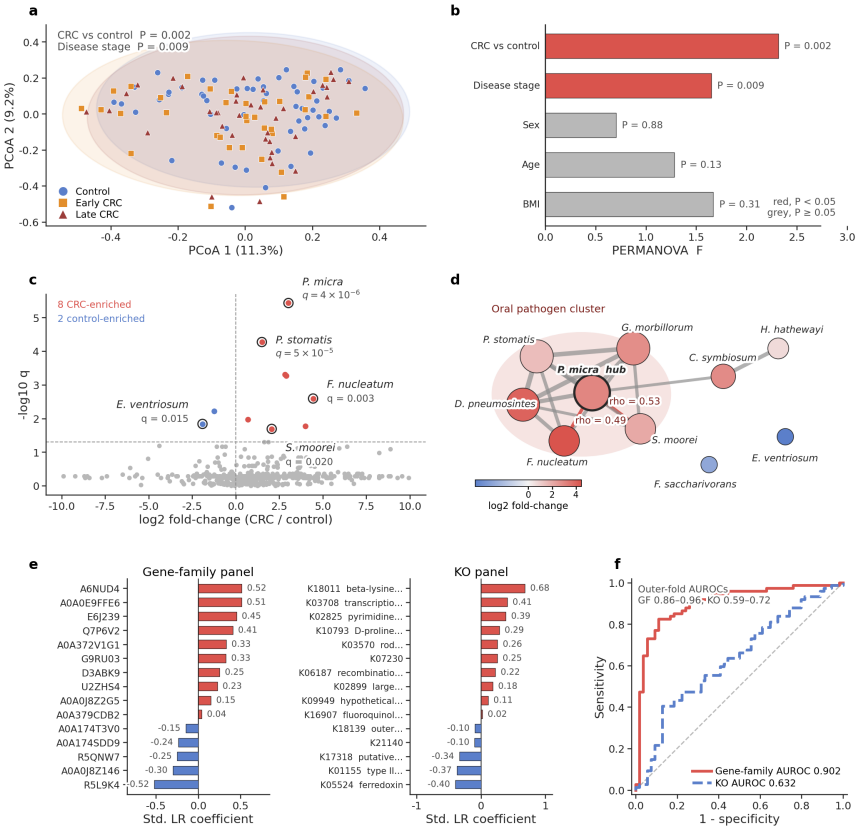



Figure 3: Reference-based analysis of CRC faecal metagenomes. Species-level beta diversity, disease-associated taxa, co-occurrence of the differentially abundant species, functional features, and nested cross-validation of gene-family and KO classifiers are shown.

### 2.2.3 A polyethylene-enrichment genome-resolved metagenomic study

We next evaluated whether MicroBioAgent could reconstruct genomes from enrichment metagenomes and use them to identify microorganisms and annotate genes associated with alkane and oxidized-derivative metabolism. The dataset comprised six Oxford Nanopore long-read metagenomes from two soil inocula (S1 and S2), each enriched on low-molecular-weight polyethylene (LMWPE), triacontane (C30) or tetracontane (C40) [14]. Assembly and binning generated 372 bins, of which 153 passed completeness and contamination filtering (100 high-quality and 53 medium-quality MAGs). Dereplication at

95% average nucleotide identity retained 75 representative genomes (50 high quality and 25 medium quality, figure 4a). These genomes were classified using GTDB-Tk with GTDB release R232, and their abundances across the enrichment cultures were estimated by read mapping with CoverM to identify the predominant microorganisms in each enrichment culture.

MAG abundance profiles showed different dominant taxa across the enrichment cultures. *Acinetobacter guillouiae* dominated S1 LMWPE and S1 C30, accounting for 64% and 67% of the reported abundance, respectively. S1 C40 was instead dominated by *Aquipseudomonas*, which accounted for 66% at the genus level (figure 4b,c). The S2 cultures showed a different composition. *Aquipseudomonas* sp003050835 was most abundant in S2 LMWPE (42%), while *Mesorhizobium_M* sp900116595 and *Aquipseudomonas moorei* were the most abundant species in S2 C30 (16% each). In S2 C40, *Mesorhizobium_M* sp900116595 and *Aquipseudomonas fluorescens_BJ* predominated at 27% and 25%, respectively. Taxonomic annotation identified the ten taxonomic groups of particular interest in the original study (Supplementary Table S2). The groups reported as *Pseudomonas* in the original study were classified as *Aquipseudomonas* in the present analysis due to differences in the reference taxonomy.

To examine the metabolic potential of the ten taxonomic groups emphasized in the original study, functional annotation was performed on one representative MAG from each group. The annotations identified genes associated with the three metabolic processes examined in the original study (figure 4d). Genes associated with alkane terminal oxidation were annotated in eight MAGs, while those associated with 2-ketone or oxidized-derivative metabolism were detected in all ten. Nine MAGs contained genes assigned to the $\beta$-oxidation module M00087. Eight MAGs contained annotations spanning all three categories, indicating that the corresponding genetic potential was distributed across multiple taxa of interest.

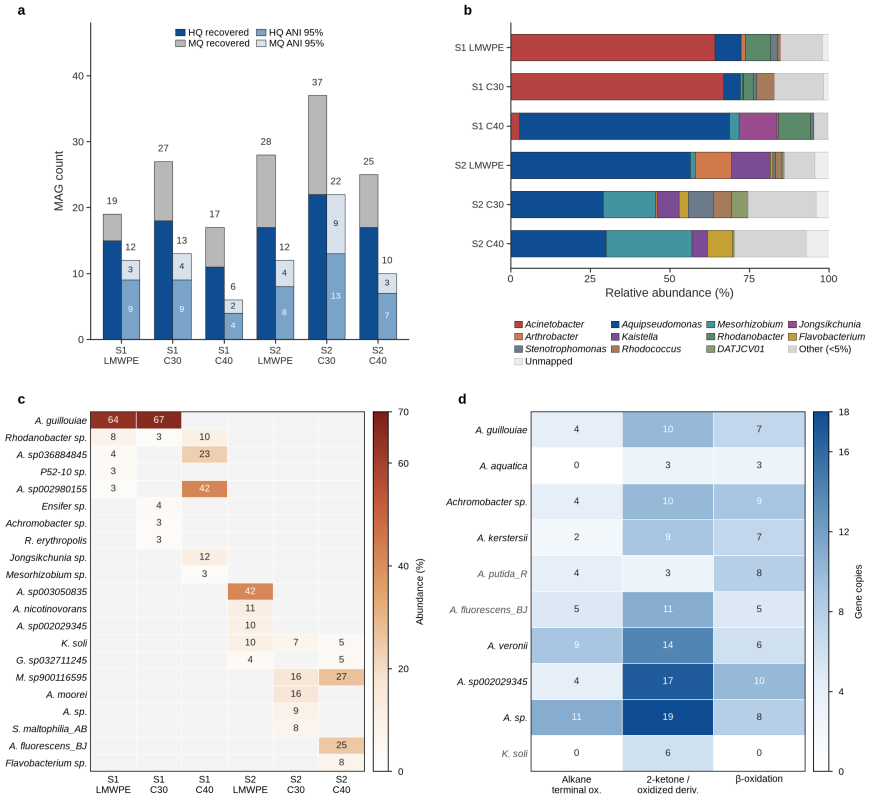



Figure 4: Assembly-based analysis of enrichment metagenomes. (a) Numbers of high- and medium-quality MAGs, grouped by completeness and contamination thresholds defined in Methods, before and after 95% ANI dereplication. (b) Genus-level MAG composition. (c) Species among the five most abundant MAGs in at least one sample. (d) Numbers of genes annotated using eggNOG-mapper in one representative MAG per target taxon for alkane terminal oxidation, 2-ketone or oxidized-derivative metabolism, and $\beta$-oxidation (M00087). Counts represent annotated genes rather than distinct functional types or transcript abundance.

## 2.3 Robustness across models and prompt granularity

Robustness will be evaluated on frozen Case 1 inputs with the complete system under three models, three prompt granularities, and three independent sessions per cell (27 sessions). The primary outcome is artifact-level completion of the common scientific goals: data characterization, 16S alpha and beta diversity, taxonomic and differential-abundance analysis, and shotgun taxonomic and differential-abundance analysis. Method-specific adherence is scored only for the detailed prompt. Shannon agreement, distance-matrix Mantel tests, and Procrustes comparisons of ordinations provide secondary repeatability measures.

Table 2: Planned reporting table for robustness across models and prompt granularity. Each cell summarizes three independent sessions. Values will be inserted after the 27-session experiment is complete.

| Model | Prompt | Completed sessions | Artifact score mean / 5 | Shannon agreement $r$ | Mantel agreement $r$ | Procrustes disparity |
|---|---|---|---|---|---|---|
| Model A | Detailed | pending | pending | pending | pending | pending |
| | Medium | pending | pending | pending | pending | pending |
| | Simple | pending | pending | pending | pending | pending |
| Model B | Detailed | pending | pending | pending | pending | pending |
| | Medium | pending | pending | pending | pending | pending |
| | Simple | pending | pending | pending | pending | pending |
| Model C | Detailed | pending | pending | pending | pending | pending |
| | Medium | pending | pending | pending | pending | pending |
| | Simple | pending | pending | pending | pending | pending |

*Notes:* Completed sessions are reported as $n/3$. The artifact score sums five prespecified analysis domains, each scored 0, 0.5, or 1. Agreement statistics summarize pairwise comparisons among repeats within a model–prompt cell. Mantel results are reported separately by distance metric in the Supplementary Information, the main-table value summarizes metrics shared by all three sessions. Lower Procrustes disparity denotes closer agreement.

## 2.4 Evaluating component contributions and asynchronous guidance

### 2.4.1 Component contributions to end-to-end analysis

End-to-end component ablation will compare the complete system with conditions lacking shipped skills, coding and long-execution guidance, or the session-runtime extension. Each condition uses the same detailed Case 1 prompt, frozen inputs, one model, and three independent sessions without mid-run user feedback. The primary outcomes are end-to-end task success, artifact completion across nine requested outputs, adherence to the eight prespecified methods, and total token use. Task completion is additionally

resolved by output type to identify which parts of the requested analysis are omitted under each ablation. Raw token use will be interpreted together with completion because a stalled session can consume fewer tokens by performing less analysis.

Table 3: Planned reporting table for end-to-end component ablation. Each condition contains three independent sessions using the same model, detailed prompt, and frozen inputs.

*A. Core ablation outcomes*

| Condition | Task completion rate (%) | Method adherence rate (%) | Median total tokens |
|---|---|---|---|
| Full | pending | pending | pending |
| −Runtime extension | pending | pending | pending |
| −Skills | pending | pending | pending |
| −Exec guidance | pending | pending | pending |

*B. Completion of individual requested tasks*

| Task | Full | −Runtime extension | −Skills | −Exec guidance |
|---|---|---|---|---|
| 16S $\alpha$ diversity | pending | pending | pending | pending |
| 16S $\beta$ diversity | pending | pending | pending | pending |
| 16S taxonomic table | pending | pending | pending | pending |
| 16S differential abundance | pending | pending | pending | pending |
| MetaPhlAn table | pending | pending | pending | pending |
| Shotgun differential abundance | pending | pending | pending | pending |
| Figures | pending | pending | pending | pending |
| Scripts | pending | pending | pending | pending |
| Report | pending | pending | pending | pending |

*Notes:* Full denotes the complete system, −Runtime extension uses the standard session runtime, −Skills removes the microbiome domain skills, and −Exec guidance removes the coding and long-execution instructions. Each task-specific cell in panel B reports the number of sessions that produced a valid result for that task ($n/3$). Task completion rate is the percentage of the nine task items completed, averaged across the three sessions in a condition. Method adherence rate is the percentage of the eight methods named in the detailed prompt: sequencing-depth normalization, Shannon diversity, weighted UniFrac, unweighted UniFrac, Bray–Curtis, weighted-UniFrac PCoA, PERMANOVA, and PERMDISP that were used as requested, averaged across the three sessions. Median total tokens use the same accounting source in all conditions. Individual-session and pre-failure token counts are reported in the Supplementary Information.

### 2.4.2 Runtime ablation for analysis continuity

We tested asynchronous guidance with a real, minutes-scale Case 1 QIIME 2 taxonomy-classification task. After the command was confirmed running, the user instructed the agent to replace a pooled resistant-versus-susceptible comparison with separate T1, T2C, and T2 comparisons and to add a significant-genera heatmap. No subsequent “continue” message was sent.

With the extended runtime, the agent yielded while the command continued, updated the pending plan during execution, and resumed automatically after command completion in all three sessions (table 4). With the standard runtime in the tested configuration, neither plan revision during active computation nor automatic session resumption after tool exit was observed in any of the three sessions. These observations support plan revision and automatic continuation for the tested command and runtime configuration. The two endpoints assess session interaction. They do not separately establish that every requested change was correctly reflected in the final analytical outputs.

Table 4: Runtime-extension ablation assessing asynchronous human guidance during active computation. Each runtime was evaluated in three independent sessions.

| System | Plan updated during execution | Automatic continuation after tool exit |
|---|---|---|
| Extended runtime | (3/3) | (3/3) |
| Standard runtime | (0/3) | (0/3) |

*Notes:* The two conditions differ only in the session-runtime extension described in section 4.1.3. Plan updated during execution requires the agent to yield while the tool continues in the background and to incorporate the user's message into the pending plan before tool completion. Automatic continuation requires tool exit to reactivate the session without another user message. Results are reported as successful sessions out of three.

# 3 Discussion

MicroBioAgent connects biological objectives to established microbiome methods through an analysis process that researchers can inspect and revise. The three reanalyses illustrate complementary uses of this approach: relating amplicon and shotgun profiles to resistance phenotypes, extending disease profiles into functional and predictive analyses, and linking taxonomically assigned MAGs to metabolic potential. Their common contribution is the composition of existing methods around a study question. The underlying algorithms and reference resources remain responsible for the primary biological estimates.

Organizing domain knowledge at the level of individual procedures provides a practical basis for this composition. Method requirements and expected products remain explicit while their arrangement can vary between studies. The case studies also show why interpretation must retain the context of these choices. A nominal association may not survive multiple-testing correction, a taxon may be represented differently across reference databases, and a predicted gene repertoire does not establish metabolic activity. Maintaining commands, parameters, and outputs within the analysis context supports review of such distinctions, although their correct interpretation

still requires scientific judgment.

The current evidence establishes feasibility across the selected cases. The planned robustness and component-ablation experiments have not yet established aggregate completion rates, model dependence, or the contribution of each component to analytical performance. Comparisons with a general agent given the same software and resources would further quantify the value of the integrated system. Agreement between repeated analyses should be assessed alongside method correctness and independently checked results, since consistent outputs can still be incorrect. User studies are also needed to determine whether the conversational interface reduces training, manual intervention, or time spent organizing an analysis.

Human confirmation of the initial plan and expert review of analytical choices remain part of the operating model. Procedure-specific output checks do not guarantee statistical validity or prevent unsupported interpretations. Local execution retains the compute, software-maintenance, and reference-database requirements of the underlying tools. Workflow managers remain well suited to repeated execution of a specified analysis at scale. MicroBioAgent focuses on defining and revising study-specific analyses through interaction.

Continuity during prolonged computation supports this operating model. The targeted runtime experiment demonstrated plan revision during an active command and automatic continuation after tool completion in the tested configuration. These observations establish a limited execution capability. reliability across longer jobs and more complex revisions, and the fidelity of final outputs to revised instructions, require further evaluation.

MicroBioAgent organizes established microbiome methods into study-specific analyses guided by biological questions. The three published-data reanalyses demonstrate paths through amplicon profiling, reference-based metagenomics, and MAG construction with taxonomic and functional characterization. By connecting modular domain procedures, local computational resources, and intermediate results within a common analysis context, the system provides a basis for question-driven analysis across microbiome workflows.

# 4 Methods

## 4.1 Details on implementing MicroBioAgent

MicroBioAgent was implemented on OpenClaw to plan and execute microbiome analyses from natural-language requests (figure 1). Users can interact with MicroBioAgent through OpenClaw's web interface, a command-line client, or supported messaging applications such as WhatsApp, Slack and Telegram. These interfaces communicate with the OpenClaw gateway, a backend service that manages conversations, language-model calls and tool

execution. The agent combines instructions for conducting an analysis with method-specific guidance and configured computational resources to carry out upstream and downstream tasks on a user-controlled workstation or server.

The behaviour of MicroBioAgent is guided by Markdown instruction files in the agent workspace. `AGENTS.md` defines how analytical tasks are planned and tracked, `TOOLS.md` describes how tools and reference resources are accessed, and `CODING.md` specifies how analysis code and long-running commands are executed. These instructions operate alongside method-specific skills, each documented in a `SKILL.md` file describing the method's applicability, input requirements and execution procedures. The agent uses this guidance to compose a study-specific plan. The `runtimes.yaml` configuration identifies available tools and their software environments, while `references.yaml` specifies reference resources and their local paths. Users can extend the system by adding or revising skills and registering additional installed tools and prepared databases.

Execution instructions and a runtime patch allow long-running analytical tasks to continue in the background while users provide feedback through the conversation. This feedback can revise the remaining analysis plan without waiting for the running task to finish. When a background task completes, the system automatically reactivates the agent to inspect the outputs and continue the remaining steps.

### 4.1.1 Analytical skills and configured execution

The skills cover 16S amplicon processing, reference-based metagenomic profiling and assembly-based genome reconstruction, together with downstream statistical analysis, functional annotation and visualization (table 1). Method-specific instructions describe the required inputs, software and reference resources, relevant parameters and expected outputs. Where provided, reference scripts serve as starting points that the agent can adapt to the study data. Additional statistical analyses can be implemented using configured libraries, as in the gene-family and KO classification analyses in the colorectal-cancer case.

The tool-use instructions in `TOOLS.md` direct the agent to resolve software environments through `runtimes.yaml` and reference paths through `references.yaml`. Commands and analysis scripts are executed through OpenClaw's `exec` tool in the designated host or container environment. Conda environments provide the software required for the corresponding tasks. If a required environment or database is unavailable, the agent reports the missing resource instead of inferring its location. Installation of additional software or large reference databases during an analysis requires an explicit user request. Upstream processing and downstream analysis can run in the user-controlled environment, while optional online annotation procedures

transmit the queries required by the corresponding public services.

The execution instructions in `CODING.md` require the agent to run and inspect individual analytical operations before proceeding to dependent work. Inspection includes the exit status, logs and expected output files, with method-specific checks defined by the relevant skill. When a stage fails, valid intermediate products can be retained and the failed stage rerun. Products affected by changed parameters or invalid upstream results must be regenerated before they are used in subsequent analyses.

### 4.1.2 Analysis planning and execution

At the start of a conversation, MicroBioAgent calls `mgx_create_job` to create or reuse the analysis workspace associated with that conversation. Input data, generated code and intermediate results are stored in this workspace and remain accessible throughout the conversation. With the workspace established, MicroBioAgent formulates an analysis plan based on the research objective, input data and requested outputs, following the instructions in `AGENTS.md`.

Once execution is authorized, MicroBioAgent consults the relevant `SKILL.md` document for each analytical step and constructs the corresponding commands or analysis code. These commands and scripts are executed through the `exec` tool, which returns the execution status and captured output to the conversation. Guided by `CODING.md`, MicroBioAgent checks the generated files before using them in subsequent analyses. Intermediate findings and user feedback inform revisions to the remaining tasks.

### 4.1.3 Human-in-the-loop interaction and automatic continuation

The long-task instructions in `CODING.md` coordinate background execution with conversational interaction. The agent initially observes an `exec` command for 30 seconds. If the tool reports that the command is still running and supplies a process-session identifier, the agent calls `sessions_yield` to end the conversational turn. The command remains under runtime management and continues in the background. Commands that finish or fail during the observation period are handled directly.

Two events can reactivate the session. A user message allows the agent to address feedback and revise pending analytical tasks while the background command continues. If the command is still active after the message has been addressed, the agent yields the conversation again. When the command exits, a completion event triggers inspection of its outputs and continuation of the remaining tasks, taking account of any changes requested during execution. The instructions favour completion-triggered continuation over repeated conversational polling.

We implemented a version-pinned patch to the OpenClaw session runtime to support this behaviour. Building on existing background-execution and session-yielding facilities, the patch modifies completion-event delivery, reactivation of the associated session and model-facing tool guidance. It also handles completion events that arrive during an active conversational turn. The runtime changes provide the event-delivery and continuation mechanisms, while the execution instructions guide the agent's use of those mechanisms. The patch pack is distributed with application instructions and a manifest recording the target OpenClaw version and commit. Complete agent instructions and analytical skills are included in the source repository.

### 4.2 Scope of the related-system comparison

Table 5 summarizes reported functional scope in related bioinformatics agents [6–11, 15–17]. The comparison concerns documented capabilities, not comparative accuracy or performance under a shared evaluation. Unconfirmed capabilities are marked as unestablished rather than absent.

Table 5: Reported capabilities of bioinformatics agents relevant to end-to-end microbiome analysis. The three microbiome columns specifically concern analysis beginning with sequencing reads.

| | Agent operation | | Microbiome analysis from reads | | | Deployment |
|---|---|---|---|---|---|---|
| Agent | Autonomous workflow execution | Human-in-the-loop | 16S amplicon | Reference-based shotgun | Assembly / MAG | On-premises analysis |
| BIA | ✓ | — | — | — | — | ✓ |
| BioChatter | — | ✓ | — | — | — | ✓ |
| AutoBA | ✓ | — | — | — | — | ✓ |
| Biomni (open-source) | ✓ | — | — | — | — | ✓ |
| BioAgents | — | — | — | — | — | — |
| M3A | ✓ | ✓ | — | — | — | ✓ |
| BioMedAgent | ✓ | — | — | — | — | ✓ |
| BioMaster | ✓ | — | — | ✓ | — | ✓ |
| MetaClaw | ✓ | — | ✓ | ✓ | — | ✓(partial) |
| MicroBioAgent | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ |

*Notes:* ✓, reported support, —, support not established from the reviewed sources, which does not demonstrate inability. Biomni denotes the open-source release. This scope comparison is not a performance benchmark.

### 4.3 Case study design

We selected three published datasets to exercise complementary analytical capabilities on plant, human, and environmental microbiomes (Supplementary Table S1). The cases were designed as capability-oriented reanalyses

rather than exact computational reproductions. Software releases, reference databases, and selected statistical procedures could therefore differ from the source publications. Before examining the reanalysis output, we identified the biological comparisons and taxa or functions emphasized by each source study. Evaluation considered whether the requested analysis was completed, whether its artifacts were usable, and whether those prespecified signals were recovered. Numerical identity with the source publication was not required.

#### 4.3.1 Case 1 — Tomato rhizosphere 16S and metagenomic analysis

The tomato dataset comprised 60 V3–V4 16S samples arranged as four cultivars, five sampling conditions, and three biological replicates [12]. A57 and HG70 were treated as susceptible cultivars, whereas HG64 and HF12 were treated as resistant. The sampling conditions were T1 (day 0), T2C and T2 (day 5, uninoculated and inoculated), and T3C and T3 (day 10, uninoculated and inoculated). Thirty-six matched shotgun libraries covered T1, T2C, and T2.

Amplicon sequence variants were obtained by DADA2 denoising and assigned taxonomy using the configured QIIME 2 classifier. Alpha diversity was calculated from an ASV table rarefied to 56,726 reads per sample. Shannon values were summarized by cultivar, time point, and inoculation status. Pooled resistant and susceptible groups were compared within each time point using two-sided Mann–Whitney tests. Community dissimilarity was evaluated using weighted and unweighted UniFrac and Bray–Curtis distances. Weighted-UniFrac principal-coordinate analysis was used for visualization, and group effects were assessed by PERMANOVA with PERMDISP used to evaluate homogeneity of multivariate dispersion.

Genus-level differential abundance was evaluated separately at each sampling condition. MaAsLin 3 models used total-sum scaling followed by logarithmic transformation and contrasted each resistant cultivar with the pooled susceptible cultivars. A genus was classified as shared nominal enrichment when its coefficient was positive and $P < 0.05$ in both the HG64 and HF12 contrasts. Benjamini–Hochberg-adjusted results were retained to distinguish nominal intersections from false-discovery-rate-supported associations.

Matched shotgun libraries were profiled at species level. MetaPhlAn3 was the primary reference-based analysis, and MetaPhlAn2 was used as a database-version sensitivity analysis for the two taxa emphasized by the source study, *Pseudomonas putida* and *Sphingopyxis unclassified*. Detection and between-phenotype abundance results were reported separately for the two database generations rather than selecting one profile as the definitive answer.

### 4.3.2 Case 2 — Colorectal cancer faecal metagenomic analysis

The colorectal-cancer analysis used the C1 cohort reported by Yu *et al.*, comprising 74 CRC cases and 54 controls [13]. Among the cases, 36 were stage I–II and 38 were stage III–IV. Species relative abundances were obtained with MetaPhlAn3, and HUMAnN3 was used to quantify gene families in UniRef90. KO abundances were obtained by summing the reads per kilobase values of UniRef90 gene families mapped to each KO.

Species-level Bray–Curtis dissimilarities were analysed by principal coordinates analysis. PERMANOVA tested disease status, cancer stage, sex, age, and body-mass index. Species, gene families, and KO groups were compared between CRC and control samples using two-sided Wilcoxon rank-sum tests with Benjamini–Hochberg correction. For the differentially abundant species, pairwise Spearman correlations were corrected for multiple testing. The displayed co-occurrence network retained associations with $|\rho| \geq 0.4$ and adjusted $q < 0.05$.

Predictive analyses were performed separately on gene-family and KO abundance tables. Within each training fold, significantly differentially abundant features were ranked by Wilcoxon significance, and the top 150 formed the candidate pool. Pairwise Spearman-correlation filtering removed redundant features at $|\rho| > 0.8$, retaining at most 30 candidates. Spearman-based minimum-redundancy maximum-relevance (mRMR) selection then retained 15 features for each feature type. Feature abundances were standardized before fitting L2-regularized logistic-regression classifiers with balanced class weights.

Performance was assessed by nested fivefold cross-validation. Feature selection and tuning of the inverse regularization strength $C$ were performed in the inner loop, using $C \in \{0.01, 0.03, 0.1, 0.3, 1.0\}$. Preprocessing, feature selection and model fitting were confined to the corresponding training data. ROC curves and pooled AUROCs were calculated from the outer-fold out-of-fold predicted probabilities. Sensitivity and specificity were reported at thresholds selected using Youden's index.

### 4.3.3 Case 3 — Polyethylene-enrichment genome-resolved metagenomic analysis

The genome-resolved analysis used six single-end Oxford Nanopore long-read metagenomes from two soil inocula enriched on low-molecular-weight polyethylene, triacontane (C30) or tetracontane (C40) [14]. Assembly was performed with myloasm through Aviary. Binning used Aviary recover with MetaBAT2, Rosella, SemiBin and VAMB, followed by bin refinement with DASTool and quality assessment with CheckM2. For the quality groups reported here, MAGs with completeness $\geq 90\%$ and contamination $\leq 5\%$ were designated high quality. Remaining MAGs with completeness $\geq 50\%$

and contamination $\leq 10\%$ were designated medium quality. Dereplication used skani with average nucleotide identity $\geq 95\%$ and aligned fraction $\geq 30\%$. Taxonomy was assigned using GTDB-Tk with GTDB release R232.

Reads were cross-mapped to genomes using CoverM, with abundance quantified as reads per kilobase per million mapped reads (RPKM). The resulting MAG abundance profiles were summarized at genus and species levels. Each inoculum–substrate combination was represented by one metagenome, and MAGs were not treated as independent biological replicates.

One representative MAG was selected for each of ten target taxa chosen from the source study. Genes were predicted using Prodigal and functionally annotated using eggNOG-mapper v2.1.13. Annotated genes were counted under three functional categories comprising alkane terminal oxidation, 2-ketone or oxidized-derivative metabolism, and the KEGG M00087 $\beta$-oxidation module. These counts describe genomic coding potential. The complete gene-to-category rules and gene-level annotations are reserved for the Supplementary Information, together with a table documenting the source-study targets, representative MAG identifiers, full taxonomic assignments and the evidence supporting their correspondence.

## 4.4 Robustness and ablation experiments

### 4.4.1 Robustness across models and prompt granularity

Robustness will be evaluated with frozen Case 1 inputs in a $3 \times 3 \times 3$ design comprising three language models, three prompt granularities, and three independent sessions per cell (27 sessions). The complete system, computational environment, skills, and staged data will be held fixed. Detailed prompts specify the metrics, statistical tests, and figure types, medium prompts specify the same analytical goals but leave method selection to the agent, simple prompts state only the dataset and biological question. Each replicate will begin in a new conversation, and methodological coaching will not be provided after the plan is confirmed.

The 16S inputs comprise a post-DADA2 ASV table, representative sequences, rooted tree, denoising statistics, and metadata. Shotgun inputs comprise the 36 matched libraries after KneadData processing and random subsampling to 20% of reads with a fixed seed of 42. Previously generated diversity, MetaPhlAn, statistical, and figure outputs are excluded.

Completion will be scored from usable artifacts for data characterization, 16S alpha diversity, 16S beta diversity, 16S taxonomy and differential abundance, and shotgun taxonomy and differential abundance. Each domain will receive 0, 0.5, or 1. Method adherence will be scored separately for detailed prompts. Agreement among the three repeats of a cell will be assessed using pairwise correlation of Shannon values, Mantel correlation of shared distance metrics, and Procrustes comparison of ordinations. With three repeats per

cell, these comparisons will be treated as an initial repeatability screen.

### 4.4.2 Component-ablation design

The component ablation will use the detailed robustness prompt, the same frozen inputs, and one fixed model. Four conditions will be evaluated in three independent sessions each: the complete system, the standard session runtime without the extension, removal of the microbiome domain procedures, and removal of the coding and long-execution guidance. No feedback will be introduced while a session is running. All other workspace instructions, software resources, and reference resources will be held constant wherever they are not the component under test.

Task completion will be scored using nine binary items: 16S alpha diversity, 16S beta diversity, a 16S taxonomic table, 16S differential abundance, a MetaPhlAn table, shotgun differential abundance, figures, analysis scripts, and an analysis report. An item will be counted only when the corresponding artifact exists and is usable. Method adherence will be scored against eight requirements in the detailed prompt: sequencing-depth normalization, Shannon diversity, weighted UniFrac, unweighted UniFrac, Bray–Curtis, weighted-UniFrac PCoA, PERMANOVA, and PERMDISP. Task and method rates will be calculated per session and averaged within condition. Total token use will be obtained from the same accounting source for every session and interpreted together with completion because an early failure can consume fewer tokens by performing less work.

### 4.4.3 Runtime-extension experiment

Asynchronous intervention was evaluated with a Case 1 QIIME 2 taxonomy-classification command selected in a pilot to run for several minutes without artificial delay. The extended runtime and the standard runtime at the same OpenClaw version were each evaluated in three new sessions. The two conditions differed only in the session-runtime extension. All other configuration and workspace instructions were held fixed. While classification was confirmed to be running, the same user instruction requested replacement of a pooled resistant-versus-susceptible comparison with separate T1, T2C, and T2 comparisons and addition of a significant-genera heatmap. No subsequent user message was sent.

A plan-update success required the message to be processed and the pending plan to be revised before the classification command exited. Success for automatic continuation required command exit to reactivate the same session within five minutes without another user message. A run in which classification ended before the intervention was delivered did not test in-run feedback and was repeated with its exclusion reason retained.

## Data availability

Raw sequencing data are available from the public repositories associated with the three source studies. Accession identifiers and sample-level mapping will be listed in the Supplementary Information. Processed abundance tables, metadata, numerical data underlying the figures, MAG annotations, and session-level robustness and ablation scores will be deposited with the article archive. Raw sequencing files and large reference databases will not be redistributed. Their sources and releases will be documented.

## Code availability

MicroBioAgent source code, domain procedures, agent instructions, analysis prompts, environment specifications, and the session-runtime extension (distributed as a version-pinned OpenClaw patch pack) are available in the project repository. Scripts used to generate the reported statistical results and figures will be included in the accompanying archive.

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# Supplementary Information

Table S1: Published datasets, sequencing-library counts, analysis tasks, and evaluation targets for the three MicroBioAgent case studies.

| Case | Sequencing libraries | Analysis route | Primary evaluation target |
|---|---|---|---|
| Tomato rhizosphere | 60 16S; 36 matched shotgun | 16S diversity, taxonomy, differential abundance; MetaPhlAn profiling | Recovery of inoculation- and resistance-associated community signals |
| CRC faecal metagenomes | 128 shotgun | MetaPhlAn3, HUMAnN3, differential analysis, nested-CV classification | Taxonomic and functional case–control analysis from reads |
| Enrichment metagenomes | 6 shotgun | MAG construction, dereplication, taxonomic assignment, abundance, eggNOG annotation | Genome-resolved taxonomic and functional characterization |

Table S2: MAG identifiers and taxonomic assignments for the enrichment metagenomes. Taxonomic assignments follow GTDB release R232.

| MAG identifier | Taxonomic assignment |
|---|---|
| *Annotation records to be supplied.* | |

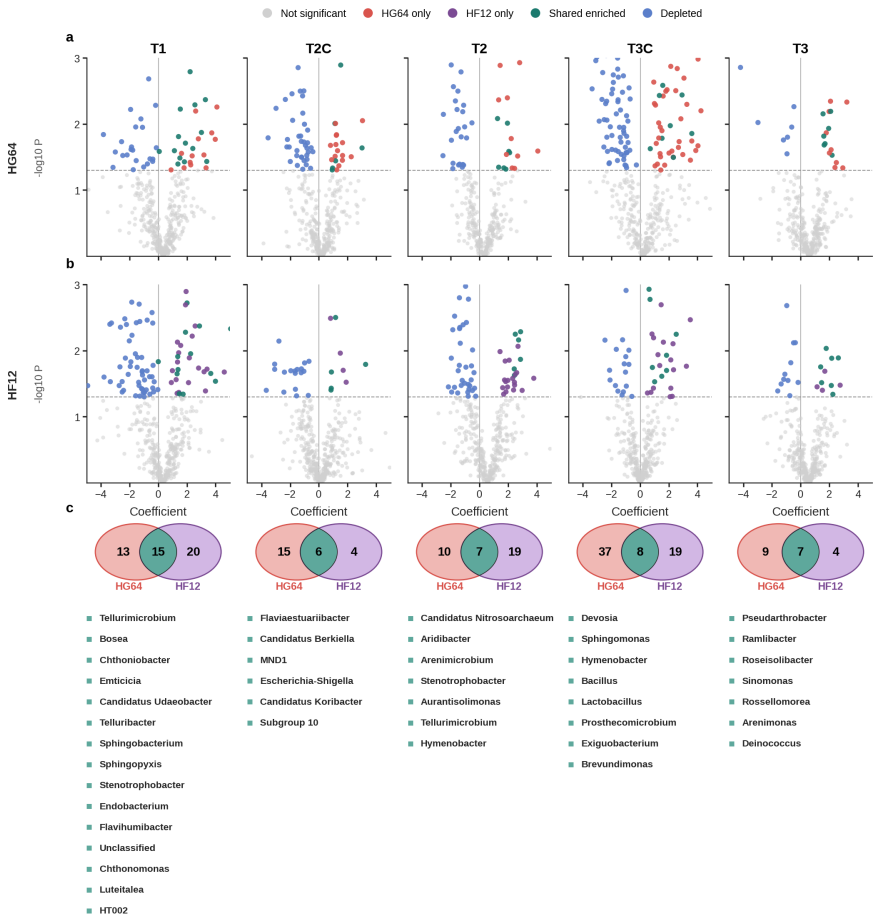



Figure S1: Differentially enriched genera in the resistant tomato parent HG64 and progeny HF12. MaAsLin 3 compared each resistant cultivar with pooled susceptible cultivars at each sampling point. Shared enrichment denotes a positive coefficient and nominal $P < 0.05$ in both contrasts.